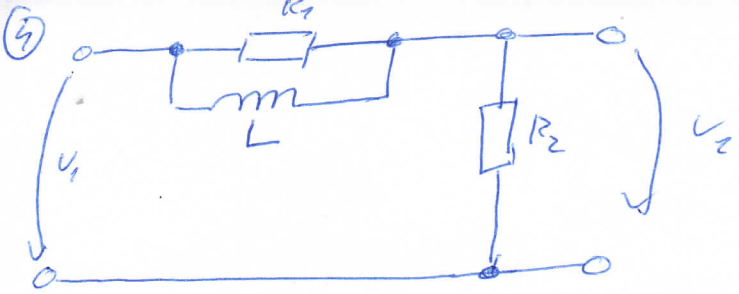
(4)

kysela



$R_1 = 8\,k\Omega$

$R_2 = 2\,k\Omega$

$L = 4\,H$

$$\boxed{\frac{R_1 \; j\omega L}{R_1 + j\omega L} = \frac{R_1}{\frac{R_1}{j\omega L} + 1}}$$

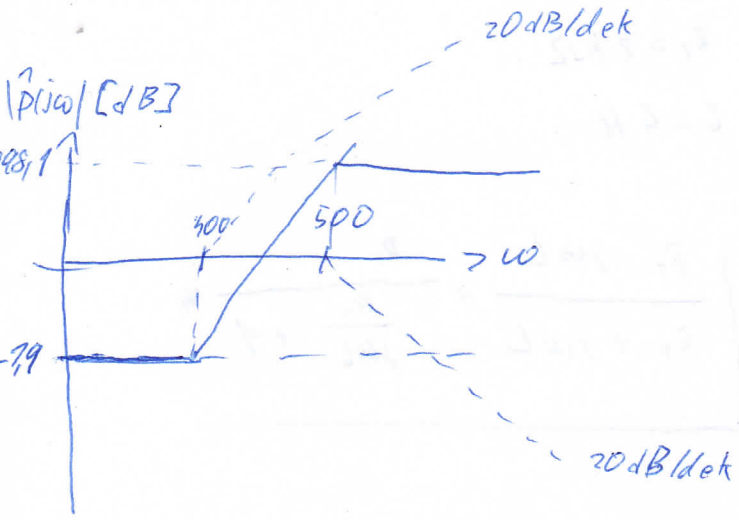
$$\hat{P}(j\omega) = \frac{R_2}{R_2 + \frac{R_1 \cdot j\omega L}{R_1 + j\omega L}} = \frac{R_2}{R_2 + \frac{R_1}{\frac{R_1}{j\omega L} + 1}} =$$

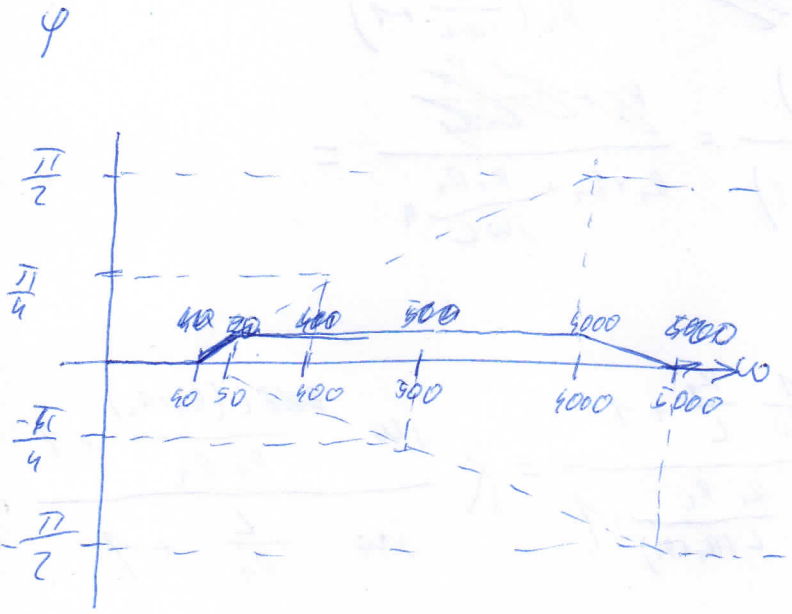
$$= \frac{R_2}{\frac{R_1 + R_2\left(\frac{R_1}{j\omega L} + 1\right)}{\frac{R_1}{j\omega L} + 1}} = \frac{R_2\left(\frac{R_1}{j\omega L} + 1\right)}{R_1 + R_2\left(\frac{R_1}{j\omega L} + 1\right)} = \frac{R_2\left(\frac{R_1}{j\omega L} + 1\right)}{R_1 + R_2 + \frac{R_1 R_2}{j\omega L}} =$$

$$= \frac{R_2}{R_1 + R_2} \cdot \frac{\frac{R_1}{j\omega L} + 1}{\frac{\frac{R_1 R_2}{j\omega L}}{R_1 + R_2} + 1} = K \frac{\frac{1}{j\omega}\frac{R_1}{L} + 1}{\frac{1}{j\omega}\frac{R_1 R_2}{L(R_1 + R_2)} + 1} = K \frac{j\omega \frac{L(R_1 + R_2)}{R_1 R_2} + 1}{j\omega \frac{L}{R_1} + 1}$$

$$K \frac{1 + j\frac{\omega}{\omega_1}}{1 + j\frac{\omega}{\omega_2}} \; ; \quad K = \frac{R_2}{R_1 + R_2} \; ; \quad \omega_1 = \frac{R_1 R_2}{L(R_1 + R_2)} \; ; \quad \omega_2 = \frac{R_1}{L}$$

$K = \frac{4}{5}$ $\qquad \omega_1 = 400$ $\qquad \omega_2 = 500$


20dB/dek
|p̂(jω)| [dB]
400
500
ω
-7,9
20dB/dek


φ
π/2
π/4
40
50
400
500
4000
5000
ω
-π/4
-π/2